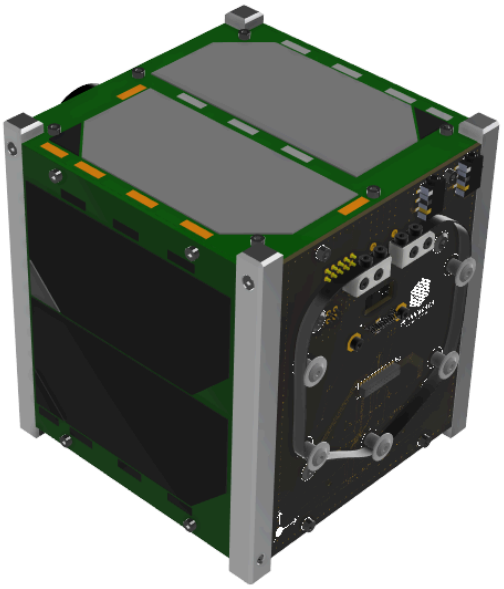
Transmission Protocol for Slippers2Sat



Prepared By

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(Communication System - Slippers2Sat)

Slippers2Sat transmits in the GFSK modulation scheme. Both Downlink and Uplink are done using this scheme. The GFSK Modulation protocol is defined below.

Communication Timing Diagram

Normal Mode	GFSK Tx (Beacon Type_1)	Receive (Telecommand Rx)	GFSK Tx (Beacon Type_2)	Digipeater Mode
	1 packet	90 s	1 packet	1 packet (90 sec)

Safe Mode	GFSK Tx (Beacon Type_1)	No Tx, No Rx	GFSK Tx (Beacon Type_2)	No Digipeater Mode
	1 packet	90 s	1 packet	90 sec

The Communication SubSystem of the Slippers2Sat has a full cycle of duration 3 minutes. The telemetry commands are accepted by the satellite after the satellite successfully transmitted a BEACON_1. In this interval of 90 seconds the satellite will receive the command from the ground station and the satellite will perform the task accordingly. After the completion of the telemetry operation the Satellite will again transmit the BEACON_2. After the complete transmission of the BEACON_2 the satellite will automatically go to the ***Digipeater MODE***. In this duration also the Ground Station can disable the ***Digipeater MODE*** and continue in the telemetry mode. Also in this time interval, the Ground Station can activate the ***Digipeater MODE*** as Ground Station operator required.

The satellite can transmit the data in every 1 second interval to the ground station. The antenna used in Satellite is a dipole antenna in UHF Amateur Band (435.313MHZ and 437.375MHz).

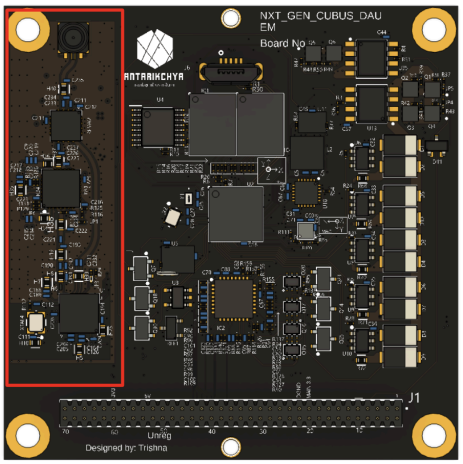

Figure 1. Daughter (BUS) Payload, showing Communication Subsystem in RED BOX

Beacon Transmission

Slippers2Sat transmits modulated Beacon with callsign and general housekeeping data of the satellite. The Beacon signal can be used to detect the presence of the satellite. It also accommodates portions of housekeeping data, antenna deployment status, mission status which is transmitted in the regular interval of 90 seconds. Baud Rate: 4800 bps

Beacon_type_1		
Information Field	**Size in bit**	**Byte**
Packet type	4 bit	2
TIM_Day	12 bit	
TIM_Hour	8 bit	1
BATT_Volts	16 bit	2
BATT_Current	16 bit	2
BATT_Temperature	16 bit	2
Raw_Current	8 bit	1
SOL_T_Volt	16 bit	2
SOL_T_Current	16 bit	2
Antenna_P_Temperature	8 bit	1
BPB_Temperature	8 bit	1
OBC_Temperature	8 bit	1
SOL_P1_Temperature	8 bit	1
SOL_P2_Temperature	8 bit	1
SOL_P3_Temperature	8 bit	1
SOL_P4_Temperature	8 bit	1
SOL_P1_Status	1 bit	1
SOL_P2_Status	1 bit	
SOL_P3_Status	1 bit	
SOL_P4_Status	1 bit	
Antenna_Dep_Status	1 bit	
MSN2_STAT	1 bit	
MSN3_STAT	1 bit	
MSN4_STAT	1 bit	
Latitude	16 bit	2
Longitude	16 bit	2
Operation_Mode	8 bit	1
OBC_RST_Count	16 bit	2
RST_Count	16 bit	2
	Total Info Bytes	31

Beacon_type_2		
Information Field	**Size in bit**	**Byte**
Packet type	4 bit	2
TIM_Day	12 bit	
SOL_P1_Volt	8-bit	1
SOL_P2_Volt	8-bit	1
SOL_P3_Volt	8-bit	1
SOL_P4_Volt	8-bit	1
SOL_P5_Volt	8-bit	1
SOL_P1_Current	8-bit	1
SOL_P2_Current	8-bit	1
SOL_P3_Current	8-bit	1
SOL_P4_Current	8-bit	1
SOL_P5_Current	8-bit	1
GYRO_X	16-bit	2
GYRO_Y	16-bit	2
GYRO_Z	16-bit	2
ACCL_X	16-bit	2
ACCL_Y	16-bit	2
ACCL_Z	16-bit	2
MAG_X	16-bit	2
MAG_Y	16-bit	2
MAG_z	16-bit	2
Last_RST	8 bit	1
	Total Info Bytes	31

GFSK Transmission

Slippers2Sat uses GFSK modulation scheme to transmit the telemetry and also in digipeater mode. The settings for the GFSK modulation scheme are as follows;
Emission Designators: 12K0F1D and 03K0F1D
EIRP: 1 dBW
Speed: 4800 bps
Packet Structure: AX.25 Packet Protocol

For Beacon

SN	Type	Length	Data Type	Value	Remark
1	Flag	1	HEX	7E	
2	Call Sign	14	ASCII		
3	Control	1	HEX	3E	
4	PID	1	HEX	F0	
5	Packet type	2	HEX	B1	
6	Information	31	HEX		
7	CRC	2	HEX		Calculate the sum from callsign to information
8	Flag	1	HEX	7E	

For Telemetry

SN	Type	Length	Data Type	Value	Remark
1	Flag	1	HEX	7E	
2	Call Sign	14	ASCII		
3	Control	1	HEX	3E	
4	PID	1	HEX	F0	
5	Packet type	1	HEX		
6	Packet Count	1	HEX		
5	Information	80	HEX		
6	CRC	2	HEX		Calculate the sum from callsign to information
7	Flag	1	HEX	7E	